

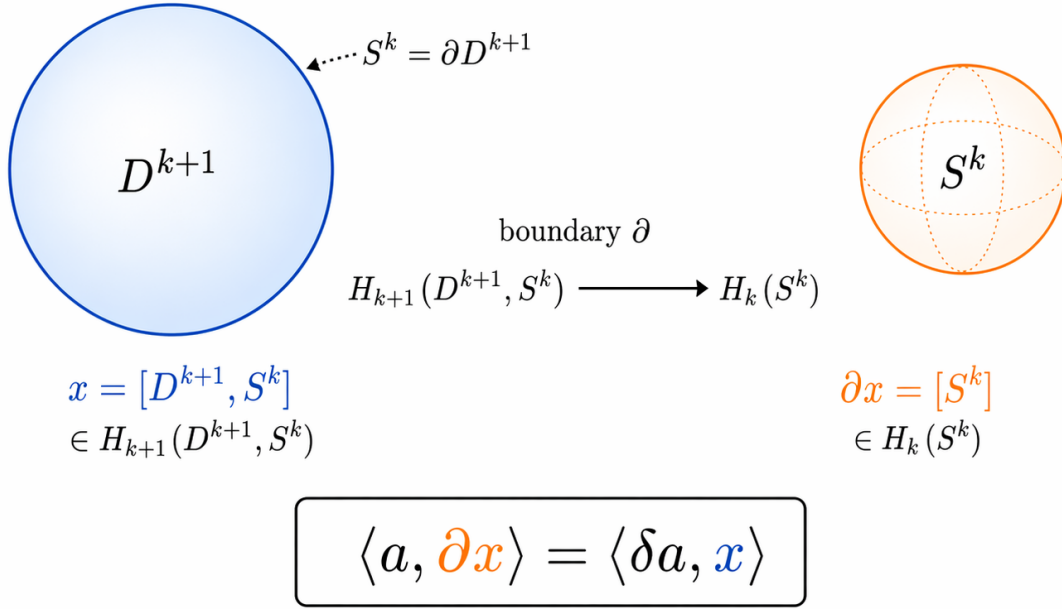
0.1 Diagrams and charts for Problems 9–10

The identity

$$\langle a, \partial x \rangle = \langle \delta a, x \rangle$$

can be understood as a relative Stokes principle: evaluating a class on the boundary of a relative cycle is the same as evaluating its connecting cohomology class on the relative cycle itself.

Pairing and boundary for (D^{k+1}, S^k)



Relative Stokes principle: evaluate on the boundary = evaluate the coboundary on the interior.

Figure 1: Pairing and boundary for the pair (D^{k+1}, S^k) .

For the annulus

$$X = S^1 \times [0, 1], \quad A = \partial X,$$

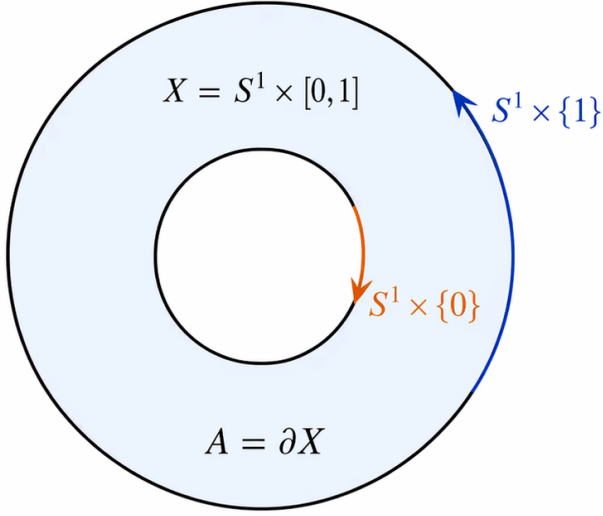
the boundary of the relative fundamental class is

$$\partial[X, A] = [S^1 \times \{1\}] - [S^1 \times \{0\}],$$

and the connecting map measures the difference of the two boundary values:

$$\delta(m, n) = m - n.$$

Annulus example: $\delta(m, n) = m - n$



$$\begin{array}{ccccc} \mathbb{Z} & \longrightarrow & \mathbb{Z} \oplus \mathbb{Z} & \longrightarrow & \mathbb{Z} \\ r & \mapsto & (r, r) & & (m, n) \mapsto m - n \\ H_2(X, A) & & H_1(A) & & H_1(X) \end{array}$$

$$\delta(m, n) = m - n$$

Relative fundamental class: $x = [X, A]$

$$\partial x = [S^1 \times \{1\}] - [S^1 \times \{0\}]$$

The connecting map measures the obstruction to extending equal boundary values into the annulus.

Figure 2: The annulus example: the connecting homomorphism detects the obstruction to extending boundary data.

For the coefficient sequence

$$0 \rightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \rightarrow \mathbb{Z}/2 \rightarrow 0,$$

the Bockstein homomorphism records the divided boundary of an integral lift. In $\mathbb{RP}^2$, one has

$$\partial e^2 = 2e^1, \quad \beta([e^2]) = [e^1].$$

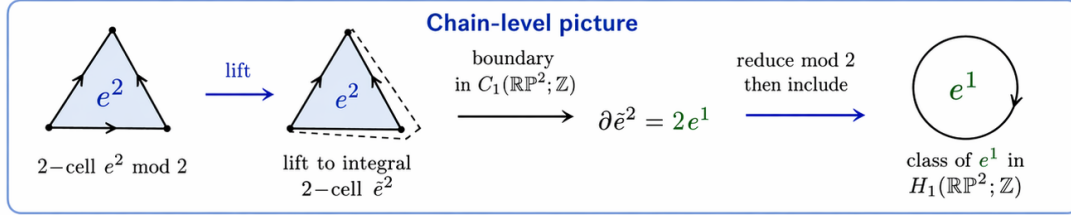
Bockstein for $\mathbb{RP}^2$

Coefficient
short exact
sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \longrightarrow \mathbb{Z}/2 \longrightarrow 0$$

Bockstein
homomorphism

$$\beta : \underbrace{H_2(\mathbb{RP}^2; \mathbb{Z}/2)}_{\text{generated by } [e^2]} \longrightarrow \underbrace{H_1(\mathbb{RP}^2; \mathbb{Z})}_{\text{generated by } [e^1]}$$



Integral cellular
chain complex
of $\mathbb{RP}^2$

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \longrightarrow 0$$

$C_2(\mathbb{RP}^2; \mathbb{Z})$ generated by $\tilde{e}^2$ $C_1(\mathbb{RP}^2; \mathbb{Z})$ generated by e^1

$$H_1(\mathbb{RP}^2; \mathbb{Z}) \cong \mathbb{Z}/2$$

generated by $[e^1]$

(since $\ker(\times 2) = \{0\}$, $\text{coker}(\times 2) \cong \mathbb{Z}/2$)

$$\beta([e^2]) = [e^1] \quad | \quad \beta \text{ is an isomorphism}$$

$$H_2(\mathbb{RP}^2; \mathbb{Z}/2) \cong \mathbb{Z}/2 \xrightarrow{\cong} H_1(\mathbb{RP}^2; \mathbb{Z}) \cong \mathbb{Z}/2$$

Figure 3: The Bockstein homomorphism for $\mathbb{RP}^2$: the mod 2 top cell detects the integral torsion class below.

Finally, the parity of n explains why the converse statement holds for even n , but fails for odd n .

Even vs odd real projective space

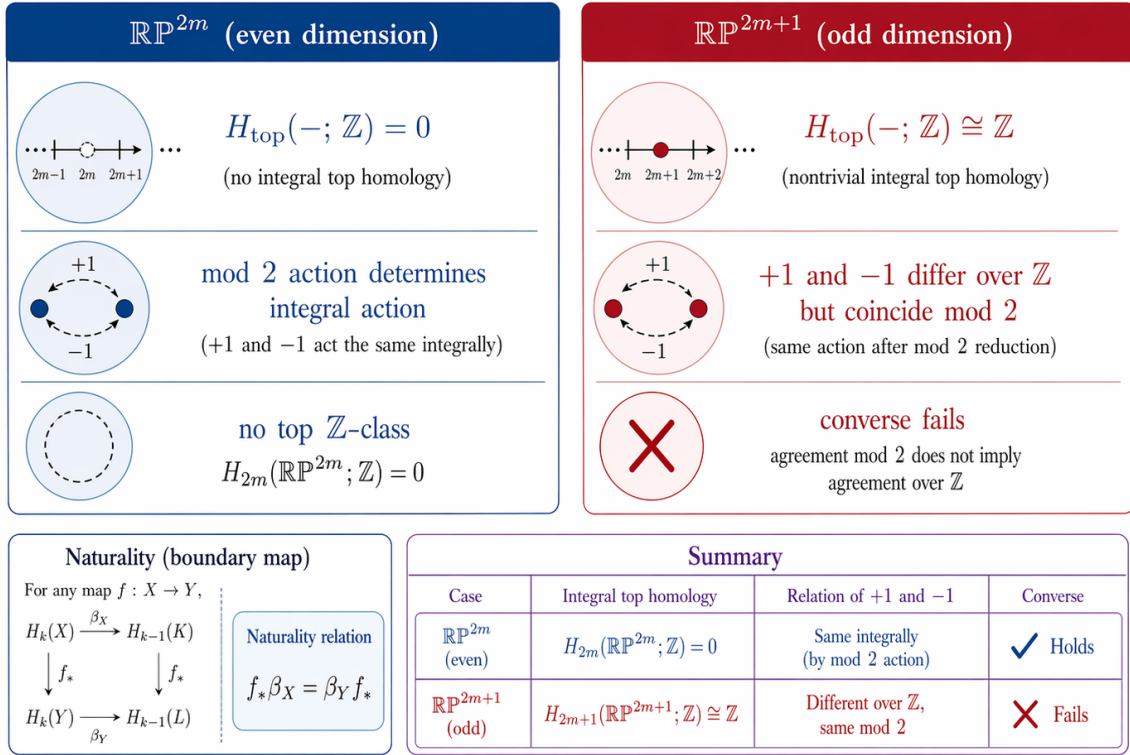


Figure 4: Even versus odd real projective spaces. Mod 2 homology cannot distinguish +1 from -1 on the top integral class when n is odd.